MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSEGEO201- CLIMATOLOGY

Duration: **2 Hr** Maximum marks:**70**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer in 60 words.
Answer any **5** questions.
Each question carries **2** marks.

1. Define black body radiation. [U] [CO1]
2. Define Precession in the Milankovitch cycle. [U] [CO2]
3. Define sensible heat transfer. [U] [CO2]
4. State what is meant by Ekman spiral. [U] [CO3]
5. List any two climatic impacts of El Nino. [U] [CO3]
6. Define tertiary circulation. [U] [CO4]

[2×5=10]

Part B

Short Notes in 250 words.
Answer any **6** questions.
Each question carries **8** marks.

7. Compare absorption, scattering and reflection of solar radiation in the atmosphere. [U] [CO1]
8. Explain temperature inversion and describe its major types. [U] [CO2]
9. Explain the origin and characteristics of trade winds. [U] [CO2]
10. Describe the formation and significance of the Ekman spiral. [U] [CO3]
11. Describe high level clouds. [U] [CO4]
12. Describe the structure and life cycle of a cyclone. [U] [CO4]

13. Summarize the salient features of the Indian monsoon system. [U] [CO5]

[8×6=48]

Part C

Essay in 400 words.

Answer any **1** Question.

Each question Carries **12** marks.

14. Discuss the planetary wind systems and explain the role of ITCZ, jet streams, and Rossby waves in global circulation. [U] [CO2]
15. Explain important precipitation mechanisms. [U] [CO4]

[12×1=12]